Sequence organisation of mother-infant interactions in chimpanzees (*Pan troglodytes schweinfurthii*) in the wild

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One fundamental feature of human communicative interactions is that they are sequentially organised, with any given turn shaped by both preceding and subsequent turns. For example, the phrase "*Sure, what time?*" can function equally as an answer to an invitation, and as a question for clarification. This sequence organisation underlies both linguistic as well as non-linguistic interactions (Kendrick et al., 2020), appears early in human development (Filipi, 2009), and is even hypothesised to precede the evolution of language itself (Levinson, 2006). As such, investigating the sequential organisation of interactions in closely-related species might shed light on features and mechanisms present at the dawn of human communication (Pika et al., 2018). Previous comparative studies have either explored aspects of sequence organisation in individually produced sequences (e.g., Girard-Buttoz et al., 2022; Peckre et al., 2025) or took a qualitative descriptive approach (e.g., Rossano, 2013; van Boekholt et al., 2024). However quantitative objective metrics of the sequence organisation of multiturn interactions are still missing. Here, we adapted methods of animal syntax research to study the sequence organisation of interactions in 17 mother-infant dyads of wild chimpanzees (*Pan troglodytes schweinfurthii*). We investigated a total of 361 communicative interactions, across four behavioural contexts (food-sharing, grooming, joint-travel, and nursing) for two components of sequence organisation: adjacency pairs, which are predictable and responsive two-turn constructs produced by two individuals, and flexibility,

defined as the variation and interchangeability of different turns. However, this variation is not random and, on an interactional level, turns or words form clusters, where unit order within a cluster is flexible, but a limited set of units makes up the cluster. As sequence organisation in humans extends beyond linguistic interactions, we took a holistic multimodal approach that included all directed behaviours between interactants. This consisted of actions (e.g. nursing) that immediately achieve their apparent goal, and signals, which require a response to achieve their goal.

To investigate adjacency pairs, a collocation analysis was conducted (Bosshard et al., 2022), a method that has successfully been used in other comparative studies looking at the non-random combination of two (or more) communicative units (e.g, Bosshard et al., 2024). The results showed that a large proportion (42%) of the 686 turn transitions were non-randomly combined into bigrams (two-unit pairs) occurring in all four investigated contexts. These bigrams were initiated by both infants (46%) as well as mothers (54%) but occurred almost exclusively (93%) unidirectionally (e.g., Raise arm → Grooming only occurred from infant to mother). This highlights that, similar to infant-caregiver interactions in humans, chimpanzees in the Ngogo community displayed an asymmetry in their interactions, where both interactants initiated different types of adjacency pairs, which could be driven by the different needs.

To investigate flexibility, we followed earlier work of Mielke and Carvalho (2022), creating a conditional transition network based on all sequentially produced signals and actions (3088 transitions). A subsequent clustering algorithm divided this network in eight highly distinguished clusters (modularity = 0.91). Seven of these clusters could be linked to one behavioural context, based on its individual signals and actions. However, one cluster, which was connected to most other clusters, contained only attention-getting signals. These results suggest that mother-infant interactions in the Ngogo community are organized in different interaction topics, where the constituent signals and actions are used with some degree of flexibility.

As this study illustrates, a quantitative, standardised approach might prove very valuable for the analysis of sequence organisation in animal communication and has the potential to facilitate direct between-species comparisons. For example, we showed that, like humans, interactions of chimpanzees of our study community are sequentially organised, strengthening the hypothesis that sequence organisation may indeed be a fundamental feature of communication, potentially predating the dawn of language.

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